Cambridge O Level Computer Science

2210 Paper 2 Practice

Algorithms, Programming and Logic

Total marks 75

Candidate name _____

Date _____

Instructions

Answer all questions. Write your answers in the spaces provided.

Use pseudocode where requested. Use standard symbols when drawing flowcharts.

The marks for each question or part question are shown in brackets [ ].

1.

Analysis is one stage of the program development life cycle.

(a) Name one other stage. **[1]**

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(b) Describe what the developer does during the analysis stage. **[3]**

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2

A user enters a sequence of numbers. The value 9999.9 signals the end of input and is excluded from the results.

(a) Write pseudocode to input the numbers and output their total. **[4]**

(b) Write a separate pseudocode algorithm to count how many of the numbers entered are greater than 100 and output the count. **[4]**

3

The algorithm below reads ten names and stores them in the array Names.

```
FOR Count ← 1 TO 10
    OUTPUT "Enter a name"
    INPUT Name
    Names[Count] ← Name
NEXT Count
```

Rewrite the algorithm in pseudocode using a condition-controlled loop in place of the FOR loop. **[4]**

4

The one-dimensional array Temperatures already contains 25 readings, starting at index 1. The values range from −20 to 100 inclusive.

Write pseudocode that uses a single loop to find the lowest reading. Output the result once, after the loop finishes. **[4]**

5

An algorithm is intended to store 50 cities and their countries in a two-dimensional array. Each row should hold the city in column 1 and its country in column 2. The algorithm should then display all 50 records.

```
01 DECLARE City : ARRAY[1:50, 1:2] OF BOOLEAN
02 DECLARE Count : INTEGER
03 DECLARE Out : INTEGER
04 Count ← 1
05 IF
06     OUTPUT "Enter the city name"
07     INPUT City[Count, 2]
08     OUTPUT "Enter the country name"
09     INPUT City[Count, 2]
10     Count ← Count + 1
11 UNTIL Count = 50
12 FOR Out ← 1 TO 1
13     OUTPUT City[Out, 1], " is in ", City[Out, 2]
14 NEXT Out
```

(a) Identify four errors in the pseudocode and give a correction for each. **[4]**

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(b) Describe how to change the corrected algorithm so the user can enter a country name and see only the cities stored for that country. **[5]**

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6

Draw a flowchart for an algorithm that adds 50 numbers. Use a counter that runs from 1 to 50. On each iteration, prompt the user for a number between 1 and 100, input it and add it to the running total. After the loop ends, display the total with a suitable message. **[6]**

7.

A school weather station records the amount of rainfall for each day of a four-week period. Rainfall is measured in millimetres and can include a decimal value, such as 3.5.

(a) Write a pseudocode declaration for a variable named Rainfall that stores one day's reading. **[2]**

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(b) Write a pseudocode declaration for a one-dimensional array named WeeklyReadings that stores all 28 readings. The first reading is stored at index 1. **[2]**

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8

A cash machine offers four options. The current balance is stored in Balance.

Choice	Action
1	Display the current balance.
2	Input a withdrawal amount. If it exceeds the balance, display "Insufficient funds". Otherwise, subtract it from the balance.
3	Input a deposit amount and add it to the balance.
4	Display "Session ended".

Assume all withdrawal and deposit amounts are positive.

(a) Write a CASE OF statement that carries out the action for a choice already stored in Choice. For any other choice, display "Unknown option". **[5]**

(b) Write pseudocode using REPEAT … UNTIL to display the menu, input Choice and execute your statement from part (a). The menu must appear at least once, and the session must end after option 4 is processed. **[3]**

9

A program stores contact details in a two-dimensional array. The user enters a first name to search for. The program checks the rows in order. When it finds the first matching first name, it displays that row's details and stops searching.

(a) Explain why the program might display the wrong contact even when the first name entered is correct. **[2]**

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(b) Explain how you could improve the search to make it more likely to find the intended contact. **[2]**

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10

A program reads positive numbers in groups. The value 0 ends a group. The value −1, entered when the program is ready to start a new group, ends the session. Every group contains at least one positive number.

(a) Draw a flowchart that inputs the numbers and displays the total and average at the end of each group. The flowchart must reset the total and count before processing the next group. **[6]**

(b) Explain why dividing the total by the count would cause a problem if the first value in a group were 0. **[2]**

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11

The array List contains 1,000 positive numbers, starting at index 1. The pseudocode is intended to find and display the largest and smallest values.

```
01 Max ← List[1]
02 Min ← List[1]
03 FOR Counter ← 2 TO 1000
04     IF List[Counter] < Max THEN
05         Max ← List[Counter]
06     ENDIF
07     IF List[Count] < Min THEN
08         Min ← List[Counter]
09     ENDWHILE
10 NEXT Counter
11 OUTPUT "Maximum: ", Max
12 OUTPUT "Minimum: ", Min
```

(a) Give one line number for each of these statement types: assignment, selection and iteration. **[3]**

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(b) Identify the line numbers of the three errors. Give a correction for each one. **[3]**

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12

A program generates 100,000 random integers from 1 to 10 inclusive. It stores the generated values in a one-dimensional array named RandomNumber. A two-dimensional array named CountedNumber stores the results. Column 1 holds the integers from 1 to 10, and column 2 holds the number of times each integer was generated.

Assume both arrays are already declared and that RANDOM(1, 10) returns a random integer in the required range.

(a) Write pseudocode using a FOR loop to generate and store all 100,000 values in RandomNumber. **[2]**

(b) Write pseudocode to set up CountedNumber and count how many times each integer appears in RandomNumber. Store each count in the corresponding row of CountedNumber. **[5]**

(c) Write pseudocode to output each integer alongside the proportion of generated values equal to that integer. Calculate the proportion by dividing its count by 100000 and round the result to four decimal places. **[3]**

End of paper